

FireGuard360

Wildfire Monitor

Empowering wildfire defense with intelligent, real-time solutions

INNOVATION CHALLENGE JUNE 2025
WILDFIRE RESILIENCE

Out Team



ADRIANO GODOY

Software Developer



DANILLO SILVA

Software Developer

PROBLEM

Wildfires are increasing in frequency and intensity due to climate change. In 2024, they caused billions of dollars in damage and forced the displacement of thousands of people worldwide.

Challenges:

- Delayed detection and response.
- Inaccurate risk prediction models.
- Lack of integrated, real-time data processing.
- Limited accessibility for non-English-speaking regions.



SOLUTION

FireGuard360 is a modular, agent-based application developed in C# (.NET 8) designed for monitoring, predicting, and responding to wildfires. It leverages a distributed architecture where specialized agents handle distinct tasks within the wildfire data processing pipeline. The system integrates with external services to enhance its capabilities, ensuring scalability, security, and real-time processing.



CORE FEATURES

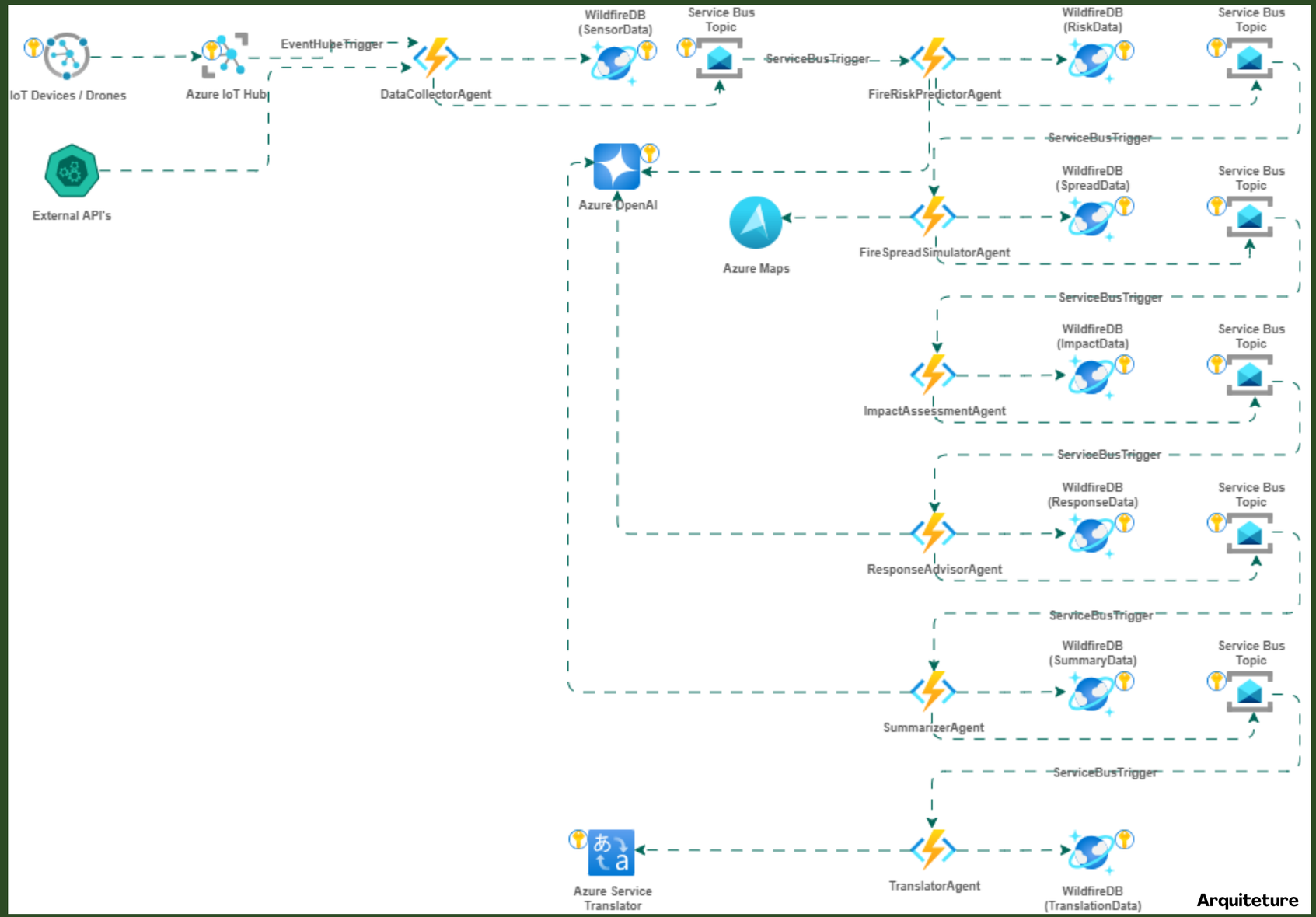
- Modular Architecture: Specialized agents for data collection, prediction, simulation, assessment, response, summarization, and translation ensure scalability.
- Real-Time Monitoring: Continuous sensor and API data ingestion for near real-time updates.
- Risk Prediction, Fire Spread Simulation and Impact analysis for infrastructure and ecosystems.
- Natural Language Summarization: AI-generated summaries for non-technical stakeholders.
- Multilingual Support.

DIFFERENTIAL

- Specialized agents for granular task efficiency.
- Seamless integration with modern cloud services.
- Accessibility for diverse global audiences.

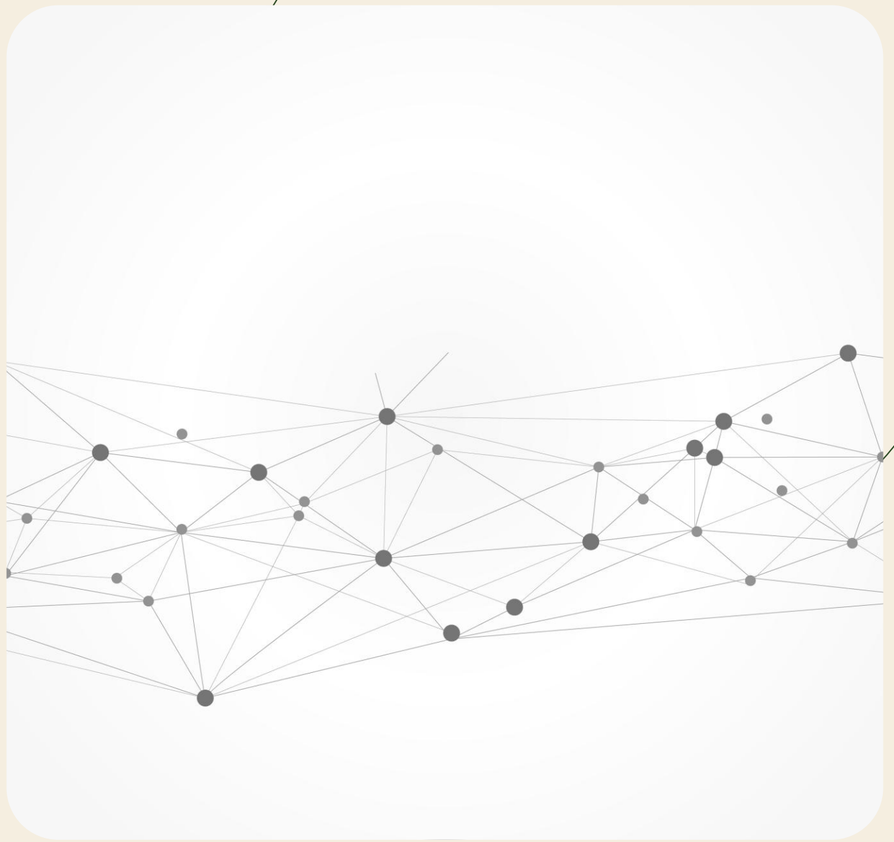
ARCHITECTURE

- Agents: DataCollectorAgent, FireRiskPredictorAgent, FireSpreadSimulatorAgent, ImpactAssessmentAgent, ResponseAdvisorAgent, SummarizerAgent, TranslatorAgent
- Asynchronous communication via Azure Service Bus
- Scalable data storage with Cosmos DB
- Secure key management with Key Vault



AGENTS

- DataCollectorAgent: Gathers data from sensors and external APIs, providing real-time environmental and situational inputs.
- FireRiskPredictorAgent: Analyzes collected data to predict areas at risk of wildfires using advanced analytics.
- FireSpreadSimulatorAgent: Simulates potential fire spread based on mathematical models, factoring in terrain, weather, and fuel conditions.
- ImpactAssessmentAgent: Evaluates the potential impact of wildfires on infrastructure, ecosystems, and communities.
- ResponseAdvisorAgent: Generates actionable response and mitigation strategies based on predictive and impact analyses.
- SummarizerAgent: Produces concise executive summaries of data and analyses for stakeholders.
- TranslatorAgent: Translates summaries and reports into multiple languages to support international collaboration.



SERVICES

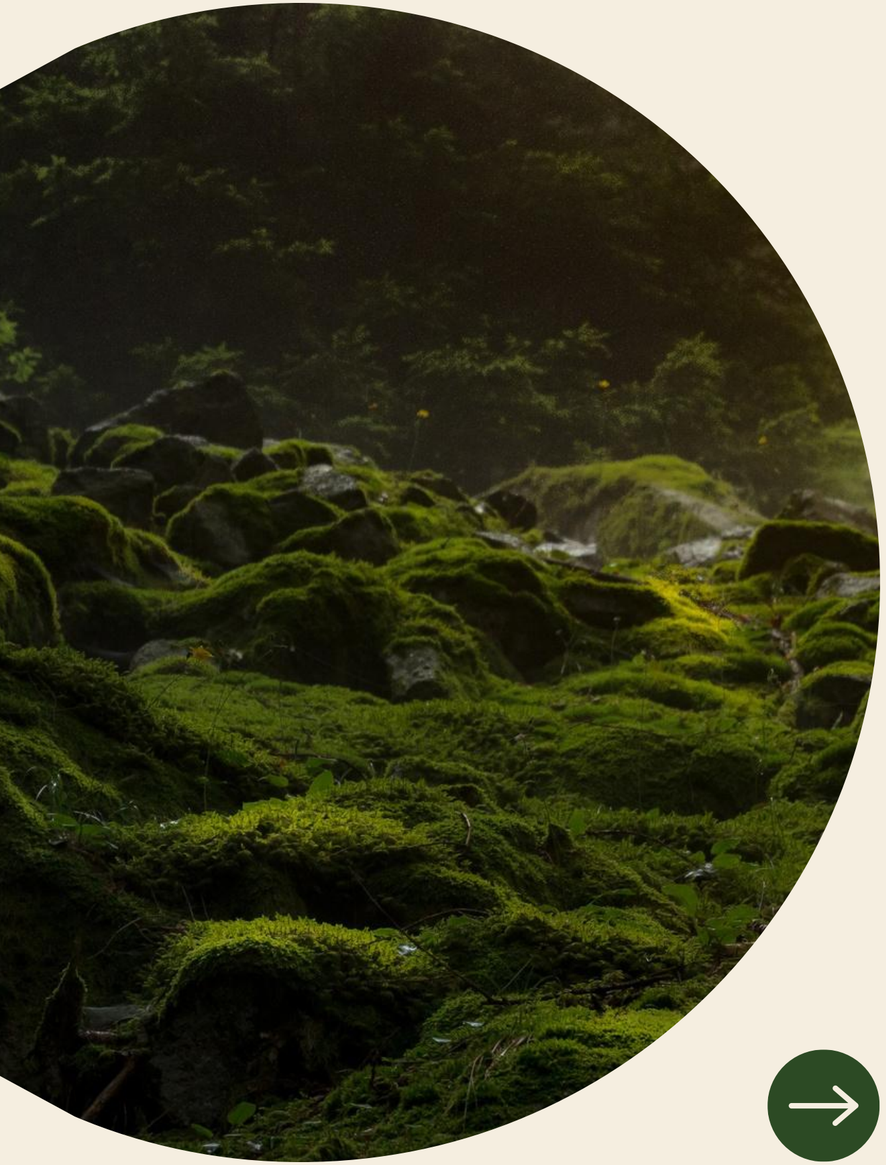
- WildfireAgents integrates with external platforms to enhance functionality:
- AzureMapsService: Provides geolocation and mapping capabilities for spatial analysis and visualization.
- CosmosDbService: Stores data in a scalable NoSQL database for efficient retrieval and querying.
- KeyVaultService: Manages secrets and keys securely to ensure system integrity.
- OpenAIService: Leverages natural language processing for enhanced data interpretation and reporting.
- Service Bus: Enables asynchronous communication between agents for seamless data flow.
- Translator Service: Supports automatic text translation to make outputs accessible in multiple languages.



WORKFLOW

1. Data Collection: The DataCollectorAgent retrieves real-time data from sensors and external APIs.
2. Risk Prediction: The FireRiskPredictorAgent processes data to identify high-risk wildfire areas.
3. Spread Simulation: The FireSpreadSimulatorAgent models potential fire spread based on environmental factors.
4. Impact Assessment: The ImpactAssessmentAgent calculates potential damages to infrastructure and ecosystems.
5. Response Recommendations: The ResponseAdvisorAgent provides actionable strategies for mitigation and response.
6. Summary Generation: The SummarizerAgent creates concise summaries for decision-makers.
7. Translation: The TranslatorAgent translates summaries into required languages for global accessibility.
8. Storage and Communication: Data and analyses are stored in Cosmos DB and shared via Service Bus for real-time updates.





RESPONSIBLE AI

1. Transparency
   - 🧠 Specialized agents (Summarizer, Predictor, Translator) with documented roles.
   - 📦 Data stored in auditable models (SummaryData, RiskData, etc.).
2. Explainability
   - 📝 Predictions and analyses come with understandable summaries and translations.
   - 🤖 Generative AI is used to explain and contextualize the results.
3. Privacy & Security
   - 🔐 Sensitive data protected via KeyVault and CosmosDb.
   - 🛡️ Access is controlled and monitored to prevent misuse.
4. Fairness & Non-Discrimination.
   - 🌐 Automatic translations ensure equal access to information.
5. Accountability
   - 👤 Automated decisions are reviewable by humans.
   - 📚 Detailed logs and continuous reviews ensure reliability.
6. Safety & Robustness
   - 🔁 Resilient system with fallback and continuous monitoring.
   - 🧩 Designed to withstand AI or infrastructure failures.



PURPOSE

WildfireAgents aims to provide a comprehensive solution for wildfire management by combining real-time data collection, predictive analytics, impact assessment, and actionable response strategies. Its modular architecture, structured data models, and external integrations make it adaptable to various environmental and operational contexts, supporting stakeholders in mitigating wildfire risks effectively.

Thank You.

FireGuard360



Contact Us

↓

https://www.linkedin.com/in/adriano-godoy-84246051/

https://www.linkedin.com/in/danillo-ara%C3%BAjo-silva-60b335124/